NAME

mbm_rollerror − Macro to read a multibeam bathymetry file, calculate the noise in the vertical reference used by the sonar, and generate a file of roll corrections for use by **mbprocess**.

VERSION

Version 5.0

SYNOPSIS

mbm_rollerror **−I***file* [**−F***format* **−W***filterwidth*]

DESCRIPTION

mbm_rollerror is a Perl macro that estimates a time series of roll bias correction values from the crosstrack seafloor slope reported by a swath sonar system, on the assumption that short−timescale variations in the apparent crosstrack slope of an otherwise smooth seafloor are caused by noise in the sonar's roll reference rather than by real topography.

The macro first runs **mblist** on the input file to extract two columns of data: the ping time (as unix/epoch seconds) and the apparent seafloor crosstrack slope (in degrees, calculated by fitting a line to the bathymetry of each ping, with positive values indicating slopes dipping to port). It then runs the GMT program **filter1d** on this time series using a cosine−arch (boxcar-like) low−pass filter of width *filterwidth* (**−Fc***filterwidth*, with the **−E** option so that filtered values are also produced near the ends of the time series) to obtain a smoothed estimate of the true, slowly varying seafloor slope. Finally, for each ping, the macro subtracts the smoothed slope from the raw slope to obtain a residual, high−frequency roll noise estimate, and writes the ping time and residual value to an output file. This output file can be applied as a roll bias correction to the survey data using **mbprocess** (see the **mbprocess** roll correction file options), or examined directly to assess the roll noise characteristics of the sonar/vessel combination.

The output correction file is named by appending the suffix ".cor" to the input filename *file*. Two temporary files (containing, respectively, the raw **mblist** time/slope output and the **filter1d** filtered output) are created during processing and removed before the macro exits.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−I *file*
Sets the input filename. This is a required argument; if it is not supplied the macro prints a diagnostic message and aborts. This program uses the **MBIO** library and will read any swath sonar format supported by **MBIO**, though useful roll noise estimates depend on the bathymetry in *file* being reasonably dense and reasonably free of a real crosstrack seafloor slope trend.

−F *format*
Sets the data format identifier for *file*. If this option is omitted, the format is determined automatically by running **mbformat** on *file*; if that automatic detection reports a format of 0 (no format recognized), *format* is instead set to −1, the **MBIO** convention indicating that *file* should be treated as a

datalist of swath data files rather than as a single swath data file. A list of the swath sonar data formats currently supported by **MBIO** and their identifier values is given in the **MBIO** manual page.

−W *filterwidth*
Sets the width, in seconds, of the cosine−arch low−pass filter (GMT **filter1d −Fc** option) used to separate the slowly varying seafloor slope trend from the higher−frequency roll noise signal. Default: *filterwidth* = 60.

EXAMPLES

Suppose we have a swath file named survey.mb59 covering a reasonably flat area of seafloor, and we suspect the sonar's roll reference is noisy. The command:

mbm_rollerror −Isurvey.mb59 −W60

will generate a roll correction file named survey.mb59.cor containing a time series of estimated roll noise values, suitable for use as a roll bias correction file with **mbprocess**.

SEE ALSO

mbsystem(1), **mblist**(1), **mbprocess**(1), **mbformat**(1)

BUGS

Maybe. Maybe not.